

Summary

- Create a new web application
- Setup development environment
- Enable React in a page
 - CDN, Offline library
- Creating virtual DOM and render into actual DOM.
 - upto 17x, 18x & 19x

React Components

- Components are reusable templates or widgets.
- A component comprises of 3 basic elements
 - a) Presentation
 - b) Styles
 - c) Functionality
- Presentation is using JSX (Javascript XML).
- Styles using CSS.
- Functionality can be defined with JavaScript or Typescript.
- You can create components in React by using
 - a) Function
 - b) Class

Javascript Function Concepts

- Function Refactor
- Declaration
- Expression
- Signature
- Function Call
- Parameters
- Actual & Formal Parameters
- Default Parameters
- Rest Parameters
- Spread Operator
- Function with return
- Function closure
- Function recursion
- Function callback
- Function Promise
- Async and await
- Function call stack
- Function Generator

- Anonymous function
- IIFE Pattern
- Arrow functions
- Debounce & Throttle
- Higher order functions
- Function currying
- Pure and Impure functions

Component using Function

1. You can configure a function with declaration or expression

allows hoisting `function Login()` `// declaration -`
`{`
`}`

not allow hoisting `const Register = function() {` `// expression - will`
`}`

2. Every component function name must start with uppercase letter.

`function login() { }` `// invalid - for component`

3. Component function can't be void type.

4. Component function must return JSX element.

```
function Login()
{
  return(
    <div>
    </div>
  )
}
```

5. Component function must return only one fraction.

```
function Login()
{
    return(
        <h2> Heading </h2>    => not valid
        <p> Para </p>
    )
}
```

6. Every JSX element must have an end token or defined with self ending.

```
<img>                                // invalid
<img> </img>
<br> </br>
<input type="text" />
```

7. JSX will not allow attributes, you must use only properties.

are attributes

```
<img src="" class="img-fluid">    // src & class
```

```
document.querySelector("img").src = "path";
document.querySelector("img").className = " "
```

```
<img src="" class="" />    // invalid in JSX
<img src="" className="" />
```

Ex:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-
width, initial-scale=1.0">
  <title>Index</title>
  <script src="./node_modules/react/umd/
react.development.js"></script>
  <script src="./node_modules/react-dom/umd/react-
dom.development.js"></script>
  <script src="./node_modules/@babel/standalone/
```

```

babel.js"></script>
  <script type="text/babel">

    function ForgotPassword(){
      return(
        <div>
          <a href="#"> Recover Password </a>
        </div>
      )
    }

    function Login(){
      return(
        <form>
          <h2>User Login</h2>
          <dl>
            <dt>User Name</dt>
            <dd><input type="text" /></
dd>

            <dt>Password</dt>
            <dd><input type="password" /
></dd>

            </dl>
            <button>Login</button>
            <ForgotPassword />
          </form>
        )
      }

    const root =
ReactDOM.createRoot(document.getElementById("root"));
    root.render(<section> <Login/> <Login/> </
section>);
  </script>
</head>
<body>
  <noscript>
    Please enable Javascript on your browser
  </noscript>
  <div id="root"></div>
</body>
</html>

```

Enable bootstrap for your project

1. Install bootstrap for project

```
> npm install bootstrap bootstrap-icons --save
```

2. Link the following files to index.html

```
<link rel="stylesheet" href="./node_modules/bootstrap-icons/font/
bootstrap-icons.css">
<link rel="stylesheet" href="./node_modules/bootstrap/dist/css/
bootstrap.css">
<script src="./node_modules/bootstrap/dist/js/bootstrap.bundle.js"> </
script>
```

Ex:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-
width, initial-scale=1.0">
  <title>Index</title>
  <!-- Bootstrap Library -->
  <link rel="stylesheet" href="./node_modules/
bootstrap-icons/font/bootstrap-icons.css">
  <link rel="stylesheet" href="./node_modules/
bootstrap/dist/css/bootstrap.css">
  <script src="./node_modules/bootstrap/dist/js/
bootstrap.bundle.js"> </script>

  <!-- React Library -->
  <script src="./node_modules/react/umd/
react.development.js"></script>
  <script src="./node_modules/react-dom/umd/react-
dom.development.js"></script>
  <script src="./node_modules/@babel/standalone/
babel.js"></script>
  <script type="text/babel">

    function ForgotPassword(){
      return(
        <div>
          <a href="#"> Recover Password </a>
        </div>
```

```

    )
  }

  function Login(){
    return(
      <form className="border alert alert-
dismissible alert-warning border-1 p-4 w-25 m-2">
        <h2>User Login</h2>
        <button className="btn btn-close"
data-bs-dismiss="alert"></button>
        <dl>
          <dt>User Name</dt>
          <dd><input type="text"
className="form-control" /></dd>
          <dt>Password</dt>
          <dd><input type="password"
className="form-control" /></dd>
        </dl>
        <button className="btn btn-
warning w-100">Login</button>
        <ForgotPassword />
      </form>
    )
  }

  const root =
ReactDOM.createRoot(document.getElementById("root"));
  root.render(<section className="container-
fluid mt-3"> <Login/> <Login/> </section>);
</script>
</head>
<body>
  <noscript>
    Please enable Javascript on your browser
  </noscript>
  <div id="root"></div>
</body>
</html>

```

